

General Principles Guiding Development of Radiation Therapy Ontology (RTO)

- The Radiation Therapy Ontology (RTO) is developed in accord with the principles of the Open Biology and Biomedical Ontology (OBO) Foundry. Please see <http://obofoundry.org/principles/fp-000-summary.html> for more details.
- RTO is intended for the open use of the medical and biomedical communities.
- RTO seeks to encompass the domain of radiation therapy, as opposed to the larger domain of radiation oncology. RTO focusses on the equipment, procedures, and physics of radiation therapy; it does not claim to describe the clinical and medical aspects of radiation oncology, tumor biology and disease physiology. However, the close linkage between the two domains is not ignored and concepts and relationships that are necessary and useful to the broader domain are included.
- RTO uses the conceptual scaffolding of the Basic Formal Ontology (BFO)¹ and the Relation Ontology².
- RTO is built using the Web Ontology Language (OWL)³
- RTO falls under the general class of “formal” ontologies. The classes and properties are meant to be fundamental and relatively general. It is understood that practical applications may need a practical ontology to be merged with RTO in order to meet the specifications and constraints of particular applications. However, the intent is that the use of RTO will relieve application implementers of significant effort and establish a uniform knowledge representation and vocabulary, which characteristics have been shown to have significant synergistic properties.
- To the extent possible, use is made of classes and properties that have been defined in other OBO Foundry Ontologies. It is understood that there can be considerable overlap between concepts in different biomedical domains, yet the overlap is not sufficiently complete to base an RTO class or property on it. In that case, a new class is implemented in RTO, but pointers to the related concept should be present (see the file `rto/CONTRIBUTING.md` for details).

prepared 8 June 2023

¹Arp, Robert, Barry Smith, and Andrew D. Spear. Building ontologies with basic formal ontology. Mit Press, 2015.

²<http://obofoundry.org/ontology/ro.html>

³<https://www.w3.org/OWL/>